

Security+ 3.2 · Secure Communication

Mestre CyberSec · Lesson 3.2 — Secure Communication (9:55) · [Assistir Vídeo no YouTube](#) ↗ · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: VPN: túnel criptografado por rede pública — IPsec na camada 3 (rede); TLS/SSL na camada 7 (app). Site-to-site VPN liga 2 escritórios. SD-WAN: software-defined — roteia por aplicação. SASE: converte tudo pro nuvem (VPN+NGFW+proxy+CASB+ZTNA).

💡 Key Exam Highlights & Critical Concepts

- IPsec vs TLS: camada 3 vs aplicação
- SD-WAN: roteamento dinâmico por aplicação e qualidade
- SASE: rede + segurança como serviço na nuvem (o novo modelo)

🎯 Exam Tips & Typical CompTIA Scenarios

- 'encrypted tunnel over public network'=VPN

Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 3.2 — Secure Communication)

Why is this advanced coverage required for SY0-701? Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Virtual Private Network (VPN)

An encrypted tunnel established over a public network (e.g., the Internet) to provide confidentiality and integrity for data in transit between remote endpoints.

 **Exam Trap:** Exam trigger: 'encrypted tunnel over public network' = VPN. Distinguish from SSL/TLS used only for application-layer encryption.

• Internet Protocol Security (IPsec)

A Layer 3 (Network layer) protocol suite that authenticates and encrypts each IP packet, commonly used for site-to-site VPNs and operating in Transport or Tunnel mode.

 **Exam Trap:** Know IPsec operates at Layer 3 (Network) while TLS/SSL operates at Layer 7 (Application). Tunnel mode is the default for VPNs.

• Transport Layer Security / Secure Sockets Layer (TLS/SSL)

Application-layer (Layer 7) cryptographic protocols that secure communications over a network, most visibly used by HTTPS to encrypt web traffic.

 **Exam Trap:** SSL is deprecated; TLS is the current standard. If the question shows a browser lock icon or HTTPS, think TLS — not IPsec.

• Site-to-Site VPN (S2S VPN)

A permanent VPN connection that links two entire networks (e.g., two branch offices) through a secure tunnel, typically using IPsec.

 **Exam Trap:** Trigger: 'connecting two offices/datacenters permanently' = Site-to-Site VPN. Contrast with client-based remote-access VPN used by individual users.

• Software-Defined Wide Area Network (SD-WAN)

A virtualized WAN architecture that uses software to dynamically route traffic across multiple transports (MPLS, broadband, LTE) based on application requirements and link quality.

 **Exam Trap:** Key distinction: SD-WAN routes by application/quality, not just destination IP. Often replaces traditional MPLS circuits.

• Secure Access Service Edge (SASE)

A cloud-delivered architecture that converges networking and security services (VPN, NGFW, SWG, CASB, ZTNA) into a single unified, cloud-native model.

 **Exam Trap:** Gatilho: 'network + security as a cloud service' = SASE. Pronunciado 'sassy'. É o modelo que substitui appliances on-premises por serviços na nuvem.

• Cloud Access Security Broker (CASB)

A security policy enforcement point placed between cloud service consumers and providers to enforce data security, compliance, and access policies.

 **Exam Trap:** CASB is a component of SASE. Trigger: 'enforcing policy on SaaS usage' (e.g., Microsoft 365, Salesforce) = CASB.

• Zero Trust Network Access (ZTNA)

A security model that requires strict identity verification and continuous authentication for every user and device before granting access to any resource, regardless of network location.

 **Exam Trap:** Core principle: 'never trust, always verify.' ZTNA replaces traditional VPN trust models. It's also a building block of SASE.

✓ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A company is merging two branch offices over broadband internet. The security team must encrypt all traffic between the two office routers, including non-web protocols, and preserve internal IP addressing. Which solution best meets this requirement?

- A) A TLS clientless VPN portal that publishes internal web applications
- B) An IPsec site-to-site VPN between the two office edge gateways
- C) A CASB configured to discover unsanctioned SaaS applications
- D) An SD-WAN policy that prioritizes VoIP traffic over MPLS

QUESTION 2 · EXAM SCENARIO

A remote user on a managed laptop needs to reach several internal applications, including SSH, RDP, and a proprietary client-server database tool, as if connected directly to the corporate LAN. The company wants strong encryption and authentication at the network layer. Which technology is most appropriate?

- A) Clientless SSL VPN
- B) CASB inline proxy
- C) IPsec remote-access VPN
- D) ZTNA broker that publishes only HTTP applications

QUESTION 3 · EXAM SCENARIO

A retail chain has 200 stores using broadband and LTE links. Network engineers need to route point-of-sale traffic over the lowest-latency path, send backup traffic over the least-cost link, and centrally manage policies by application instead of by physical circuit. Which technology best fits?

- A) SD-WAN
- B) IPsec transport mode
- C) A CASB with API-based cloud app discovery
- D) TLS mutual authentication

QUESTION 4 · EXAM SCENARIO

A global company wants to provide remote workers and branch offices with secure web gateway, firewall, CASB, and zero trust access to private applications delivered from the cloud. The security team prefers a single cloud-native architecture rather than backhauling all traffic to a data center. Which architecture should be selected?

- A) SASE
- B) A traditional hub-and-spoke site-to-site VPN architecture
- C) An SD-WAN overlay that provides local internet breakout but no integrated security services
- D) A client-based full-tunnel IPsec VPN that sends all remote traffic to the data center

QUESTION 5 · EXAM SCENARIO

Employees are using unsanctioned cloud storage and collaboration apps from corporate devices. The security team must discover which SaaS apps are in use, classify data uploaded to them, block high-risk apps, and enforce DLP policies without deploying a separate appliance at each branch. Which technology is designed for this?

- A) CASB
- B) IPsec tunnel mode
- C) SD-WAN forward error correction
- D) TLS certificate pinning

QUESTION 6 · EXAM SCENARIO

A company allows contractors to access an internal billing application but not the rest of the corporate network. Security requires continuous identity and device posture checks for each session, with per-application access instead of broad VPN connectivity. Which solution best meets this?

- A) ZTNA
- B) Site-to-site IPsec VPN
- C) SSL offloading on the load balancer
- D) SD-WAN application-aware routing

QUESTION 7 · EXAM SCENARIO

An organization's public web application must protect data in transit between customers and the web server, and clients must verify the server identity using a trusted certificate authority. Which protocol provides this protection?

- A) IPsec AH
- B) TLS
- C) SFTP
- D) SNMPv3

QUESTION 8 · EXAM SCENARIO

A network architect is configuring an IPsec VPN between two firewalls. The requirement is to protect the entire original IP packet, including source and destination addresses, as it crosses the public internet. Which IPsec mode should be used?

- A) Transport mode
- B) Tunnel mode
- C) Aggressive mode
- D) Main mode

QUESTION 9 · EXAM SCENARIO

A company wants to let business partners access a single internal web-based procurement portal through a browser. Partners should not receive a network-layer tunnel or access to any other internal resource. Which VPN capability is most appropriate?

- A)** Clientless SSL/TLS VPN
- B)** Full-tunnel IPsec VPN
- C)** Split-tunnel IPsec VPN
- D)** SD-WAN with dynamic multipoint VPN

QUESTION 10 · EXAM SCENARIO

A hospital's remote clinicians need secure access to SaaS EHR and on-premises imaging apps. The security team wants cloud-delivered web security, cloud app control, firewall inspection, and per-application zero trust access, while avoiding backhauling remote traffic to the data center. Which architecture most directly meets these requirements?

- A)** A single office MPLS circuit with IPsec transport mode
- B)** A SASE architecture delivering cloud-based SWG, CASB, NGFW, and ZTNA services
- C)** A clientless SSL VPN only, hosted on a data center appliance
- D)** An SD-WAN appliance that only performs link selection and QoS

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option B — An IPsec site-to-site VPN between the two office edge gateways

An IPsec site-to-site VPN operates at Layer 3 and creates an encrypted tunnel between gateways, protecting all IP traffic between the offices while preserving internal addressing. A clientless TLS VPN only publishes web applications, a CASB governs cloud app usage, and SD-WAN prioritizes or routes traffic but does not by itself encrypt all office-to-office traffic.

Question 2: Option C — IPsec remote-access VPN

An IPsec remote-access VPN provides a network-layer tunnel that can carry arbitrary IP protocols such as SSH, RDP, and database client traffic. Clientless SSL VPNs and HTTP-only ZTNA brokers are limited to web or specific application access, while a CASB controls cloud service usage rather than providing full network-layer remote access.

Question 3: Option A — SD-WAN

SD-WAN uses software-defined, application-aware policies to select WAN paths dynamically based on latency, cost, and performance, which matches the requirement to manage by application across broadband and LTE. IPsec transport mode is an encapsulation mode, a CASB governs cloud apps, and TLS mutual authentication secures application sessions but does not provide WAN path selection.

Question 4: Option A — SASE

SASE converges networking and security services such as secure web gateway, next-generation firewall, CASB, and ZTNA into a cloud-delivered architecture, which avoids backhauling remote traffic to a data center. Hub-and-spoke VPN and full-tunnel IPsec VPN backhaul traffic, while SD-WAN without integrated security services does not provide the required cloud-delivered protections.

Question 5: Option A — CASB

A CASB provides visibility into cloud service usage and enforces policies such as data loss prevention, app risk classification, and blocking of unsanctioned SaaS, often through APIs or inline proxies. IPsec tunnel mode encrypts network traffic, SD-WAN error correction improves link reliability, and certificate pinning strengthens certificate validation but does not govern cloud applications.

Question 6: Option A — ZTNA

ZTNA grants access to specific applications based on identity, device posture, and context, rather than placing users on the corporate network. Site-to-site IPsec connects offices, SSL offloading only decrypts and re-encrypts web traffic at a device, and SD-WAN routing does not enforce per-application zero trust access.

Question 7: Option B — TLS

TLS provides encryption, integrity, and certificate-based server authentication for application traffic, and it is the protocol that secures HTTPS. IPsec AH provides integrity but not confidentiality, SFTP protects file transfers rather than general web sessions, and SNMPv3 secures network management traffic.

Question 8: Option B — Tunnel mode

IPsec tunnel mode encapsulates the entire original IP packet inside a new IP packet, which is the normal choice for gateway-to-gateway VPNs. Transport mode protects only the payload of the original packet, while aggressive mode and main mode are IKE negotiation modes, not IPsec encapsulation modes.

Question 9: Option A — Clientless SSL/TLS VPN

A clientless SSL/TLS VPN publishes specific internal web applications through a browser without installing a full tunnel client or granting network-layer access. Full-tunnel and split-tunnel IPsec VPNs give network-level connectivity, and SD-WAN with DMVPN is a WAN topology and routing solution, not a browser-based limited-access portal.

Question 10: Option B — A SASE architecture delivering cloud-based SWG, CASB, NGFW, and ZTNA services

SASE is a cloud-delivered architecture that combines networking and security services such as secure web gateway, CASB, next-generation firewall, and ZTNA, enabling secure local internet breakout and per-application access without data center

backhaul. An MPLS circuit with IPsec transport mode is not cloud-delivered, a clientless SSL VPN is limited and data-center-hosted, and link selection with QoS alone does not provide the required security services.